

SMtrack v2 Backend — Tech Stack & System Architecture

1. System Overview

A backend for ingesting, processing, and storing IoT telemetry/log data (medical refrigerators/freezers with temperature/humidity/door probes) and raising alerts when readings go out of range. MQTT is the primary ingestion channel; a REST API + SSE stream serve the dashboard, and FCM delivers push notifications to mobile.

2. Tech Stack

Category	Actual technology in use
Framework	NestJS v11 (<code>@nestjs/core</code> , <code>common</code> , <code>platform-express</code> , <code>microservices</code> , <code>config</code> , <code>event-emitter</code> , <code>schedule</code> , <code>swagger</code> , <code>terminus</code> , <code>passport</code>) — hybrid app: HTTP + MQTT microservice + RabbitMQ microservice x2
ORM / DB	Prisma v7.8 (new <code>prisma-client</code> generator, output to <code>src/generated/prisma</code>) + <code>@prisma/adapters-pg</code> (driver adapter) + raw <code>pg</code> v8 for specific tasks (e.g. <code>pg-copy-streams</code> for bulk export) on PostgreSQL + TimescaleDB extension
MQTT	<code>mqtt</code> v5.10
Message queue	RabbitMQ via <code>amqplib</code> + <code>amqp-connection-manager</code> (used through <code>@nestjs/microservices</code> RMQ transport) — split into 2 microservices on separate queues (general events vs. high-volume log/telemetry)
Cache	Redis via <code>ioredis</code>
Push notification	<code>firebase-admin</code> v13 (FCM)
Auth	<code>passport</code> + <code>passport-jwt</code> — two separate schemes : <code>jwt</code> (user login) and <code>device</code> (device token)
Object storage	<code>@aws-sdk/client-s3</code> + <code>@aws-sdk/lib-storage</code> — S3-compatible (MinIO in local dev), 3 independent configs (archive, device image, firmware)
Observability	Full OpenTelemetry SDK (<code>sdk-node</code> , auto-instrumentations, OTLP HTTP exporter for both trace and metrics) + <code>nestjs-pino</code> / <code>pino</code> / <code>pino-http</code> for structured logging
Validation	<code>class-validator</code> + <code>class-transformer</code>
Testing	Jest v29 + <code>ts-jest</code> + <code>supertest</code> + <code>@nestjs/testing</code>
Other	<code>mongodb</code> v7.5 present in devDependencies (no actual usage found elsewhere in the code)